

RESEARCH REPORT · 8 . SEPTEMBER 2026

Moats and edge regeneration in quantitative retail

Trading

Systematic evidence-based scoping report for a small AI-supported trading company · Research status September 8, 2026

0 Terms and research framework

Edge here refers to a source of positive expected trading returns after relevant trading costs. Moat denotes their durability and defensibility against competition, imitation, capacity erosion and market adaptation. Meta-Edge refers to the ability to repeat

to discover new edges, validate them reliably and deploy them in a timely manner. These three properties must be proven separately.

A good backtest alone doesn't prove any of these.

The deliberately broad edge definition also includes risk premiums. Therefore, there is also a distinction between positive net return and alpha

compared to an appropriate risk and opportunity cost benchmark. An accessible risk premium can

exist for a long time without providing the individual dealer with a moat. There is no high return through leverage or rare catastrophe risks

sufficient proof of skill. Business profitability also requires that data, research, operational and capital costs be covered become; these are not implicitly equated with transaction costs.

Question 1 is therefore: Which barriers protect an edge that can actually be traded, and which ones can be reached here? H1 is: Is repeated

Edge generation measurable, economically valuable and difficult to imitate functionally? In the report, a lead over is considered "relative".

other retail players; According to the mandate, "absolute" means an advantage even over competent institutions in the field considered niche. This does not mean superiority over all institutions and markets. These comparison groups are part of the Operationalization, not a standardized academic moat classification.

Explicit default assumptions

Dimension assumption and justification

Company working title "Retail Quant Lab", proprietary trading by one person or a team of two to three people; no external fund sales organization.

Capital Small to medium capital base without a fixed euro amount. Capacity is measured relative to order size, market depth and liquidity; a blanket threshold would be unfounded.

Markets Liquid exchange-traded stocks, ETFs and futures as the starting universe. The literature is particularly stock-heavy. Illiquid securities are a niche hypothesis that must be tested separately.

Data Public information and non-exclusive, regularly purchased histories; no privileged feeds. "Public" does not necessarily mean free or already correctly historicized.

Resources Limited research and computing budget; Standard ML is available. Specific current provider prices or product comparisons are not the subject of the report.

Costs commissions, effective spread, slippage/impact, opportunity costs of unexecuted orders, if applicable, lending, financing, funding and roll costs. Personal taxes remain due to lack of jurisdiction separately.

Target Positive risk-adjusted live net performance and positive corporate surplus with a pre-determined loss budget; no promised target Sharpe.

Horizon 15 minutes, several hours and one to several days. No HFT and no transfer of monthly factors into short-term trading signals without a new test.

Crypto additionally requires exchange, custody, fragmentation and funding assumptions, FX a clear trading venue and Instrument model. The equity-based results are not applied to these markets as empirically confirmed. The report doesn't make any decisions

Statement on the legal admissibility of a specific business model.

Evidence and search method

This is a systematically structured scoping research with critical synthesis, not an exhaustive PRISMA review and not recalculated meta-analysis. The search was carried out on September 8, 2026 in publicly accessible web indices, original publisher offers, NBER,

Federal Reserve, university and author archives, and NeurIPS proceedings. The search families included "sustained competitive

advantage", "dynamic capabilities", "alpha decay", "performance persistence", "scale and skill", "backtest overfitting", "intraday predictability",

"retail limit orders" and "data-enabled learning", each combined with the relevant authors and original titles.

The first wave of searches covered theory, financial economics and ML. The second specifically looked for contrary findings: international deviations

Alpha decay, positive replications, methodological criticism of skill testing and current direct retail execution evidence.

Versions were after

work merged; historical working paper figures were not mixed with later journal results. were reasons for admission an identifiable original contribution, a verifiable reference to a key question and visible evidence. Marketing promises,

untestable

Performance reports and forums were not included as evidence of effectiveness. Search hit numbers and exclusion numbers were not

fully logged; a quantitative PRISMA flowchart would therefore be misleading.

Tier 1 includes peer-reviewed research and academic monographs, Tier 2 working papers and institutional research, Tier 3 methodologically relevant practitioner contributions, Tier 4 forums. Peer-reviewed replications are in Tier 1; their replication function becomes

additionally designated. Helmer is listed as a practitioner monograph in Tier 3. Sculley et al. are peer-reviewed conference research and are published as

Tier 1 marked, with low direct financial market transferability. The ranking does not replace an examination of the study design.

Strong means reliable for the specific statement being examined, moderately limited or indirectly supported, weak predominantly

Theory transfer or lack of direct tests. "Anecdotal" has no confirmatory weight. A theory article can be a logical one

Strongly justify the distinction without empirically proving a retail moat. All suggested actions and moat assignments are our own

Synthesis, unless a direct finding is mentioned. Full text, author version and abstract/preview access are in the bibliography differentiated.

1 Executive summary

The reviewed literature does not provide a general, reliable moat for retail ML trading between 15 minutes and several days proven; This is a limit of evidence, not proof of impossibility. The most plausible achievable advantage combines a close understanding of the market,

correct data, selective execution, limited capacity and disciplined validation; its defensibility remains to be tested

[moderate as a mechanism, weak as direct retail moat evidence]. Strategy theory requires economic benefits and

Barriers to imitation or substitution, not just effort [strong theoretical foundation; Barney 1991]. Historical

Anomalies erode on average after publication, but neither completely nor the same everywhere [strongly for examined samples;

McLean/Pontiff 2016, Jacobs/Müller 2020].

Skill exists in subpopulations according to several studies, but return persistence does not identify ongoing reinvention of alpha

[moderate and methodologically controversial; Kosowski et al. 2006, Fama/French 2010]. Direct retail evidence shows a small historical

Successfully predictable day trader group and achievable execution advantages of patient limit orders [moderate for transfer;

Barber et al. 2014, Anand et al. 2026]. Small capital base can limit impact, but does not exclude institutional competition

[moderate as a scaling mechanism, weak as a defense barrier]. ML can improve forecasts; a public

This means that a well-known model does not yet have any exclusivity [moderate; Gu et al. 2020 plus own strategy transfer].

Overall verdict H1: partially supported, with moderate confidence. Edge Regeneration is the most compelling meta ability that

Data, knowledge and process benefits activated; as an independent moat it is only plausible under additional barriers to imitation [moderate

theoretical support]. For the retail standard case described, their long-term defensibility is not supported, with weak direct defense

Data situation. The claim that every active trading must constantly find completely new edges is also too strong: monitoring, adjustment and

recurring structural sources of return may be sufficient. A prospective, realistic research register is therefore a priority

Cost measurement, small live validations and proof of additional net returns compared to a frozen comparison strategy. First

Repeated economic successes and a separately proven disadvantage of reproduction or substitution justify the name Moat.

Sources for this summary: Barney, McLean and Pontiff, Jacobs and Müller, Kosowski et al., Fama and French, Barber et al., Anand et

al., Gu et al., Eisenhardt and Martin.

2 Question 1 and the theoretical basis of the Moat

Competitive advantage and market frictions

Barney's Resource-Based View provides the test questions based on value, rarity, difficulty of imitation and lack of equivalent substitutes. A

A self-written model can contain unique code whose economic function is still easy to replicate. Vice versa

Freely accessible individual components can come together to form a combination that is difficult to reproduce. [Strong as theoretical

criterion; no direct retail test.] Barney 1991.

Porter's Five Forces help analyze the distribution of economic surplus: rivalry by competing capital, entry

through new trading teams, supplier power of data providers/brokers/exchanges, substitutes for the same return exposure and customer power

with external asset management. When it comes to pure proprietary trading, the classic end customer relationship is missing.

Sales advantage and forecast advantage

therefore must not be mixed. [Moderate as an analytical transfer.] Porter and the Institute for Strategy and Competitiveness.

Limits to Arbitrage explain why mispricing can persist despite professional competition: risky interim losses and limited funding prevents their immediate elimination. For retailers, this is both an opportunity and a source of costs; own patient capital does not eliminate margin, liquidity or loss risks. [Strong theoretical foundation, weak direct retail transfer.] Shleifer and Vishny 1997.

All seven powers in proprietary trading

The following transfer is our own analysis with weak direct empirical evidence. Helmer's categories are a useful one Practitioner framework, not an empirically exhaustive classification of trading moats. Helmer and Strategy Capital.

Power transferable mechanism and limit

Scale Economies distribute fixed costs of data and research; In retail, impact and limited capacity work against this.

Power transferable mechanism and limit

Network Economies Data or relationship networks can help. More traders of the same signal does not automatically improve its value and can reduce it.

Counter-positioning Small niches can be unattractive for large business models. Low capacity does not prove a conflict that prevents institutional imitation.

Switching Costs Relevant for fund sales or platform customers; no immediate protection of proprietary trading alpha. Your own switching costs to the broker are a dependency.

Branding Can facilitate access to capital and talent; Market prices do not reward brand names.

Cornered Resource Particularly suitable for truly exclusive data, rights or expertise; The fixed retail assumptions limit this access.

Process Power Conceivable with historically grown, economically superior and difficult to replace interaction between research and operations. A standard pipeline is not enough.

Taxonomy of possible moat sources

Reading rule: The horizons indicate possible fields of application, not an empirically proven duration of protection. There is no category for any category

verified generic half-life. "Relatively possible" requires a positive net live effect. Full sources are available in Section 11.

Moat Source Mechanism Protected

horizon

durability and

erosionStrength of Evidence Retail Accessibility Relative versus absolute

Moat

Main sources

Data Exclusive relevant histories,

correct timing, difficult

reproducible labels

15 minutes to days, each

Data latency

Until reconstruction, sale

to others or

loss of information; none

Rate known

Tier 1: moderate

theoretical; Retail

weak

data quality high,

Exclusivity low to

medium

Relatively possible;

institutional only

special

Access/Usefulness

Barney;

Hagiu/Wright

Talent and team combination

Market understanding, statistics and

Engineering

All emigration,

key person risk,

outdated knowledge

Tier 1: moderately indirect personal

Specialization

attainable; wide

Top team limited

Relatively plausible; absolutely
open
Grant;
Cohen/Levinthal;
Kosowski et al.
technology and
Infrastructure
Reliable workmanship and
Execution, low
Error costs
Especially 15
min/hours;
Operation all
Standardization and
Arming the competition
Tier 1: moderate for
differences in execution;
Retail moat weak
Reliable operation
attainable;
privileged
Infrastructure
excluded
Mostly relative competence;
narrow increase in execution
possible
Anand et al. 2012;
Sculley et al.
Process and culture Selective validation,
Mistake learning, clear
All routines can
be copyable; Path knowledge
Tier 1: moderate
theoretical, weak
Cheers for
basic discipline;
Relatively plausible; absolutely
only with imitation barrier
Teece et al.;
Eisenhardt/Martin;
Moat Source Mechanism Protected
horizon
durability and
erosion
Strength of Evidence Retail Accessibility Relative versus absolute
Moat
Main sources
Shutdown rules can protect direct superiority must
be proven
Argote
Capital and scale fixed cost coverage,
Diversification, financing
of research and
Loss phases
All, financing
especially days
Own and industry-wide
Capacity erosion
Tier 1: moderate; Theory
plus empirical
Scale findings
Limited Institutional Advantage
more common; patient

Equity as a partial advantage
Berk/Green;
Pastor et al.;
Shleifer/Vishny
smallness and
Capacity
Use small orders
limited opportunities
lower own impact
Everyone, strong
niche dependent
Other small players
or automated
institutional bundling
Tier 1: moderate for
cost/capacity;
weak as a barrier
High but illiquid
costs
Niche advantage possible;
no exclusive right to
small trades
Novy
Marx/Velikov;
Anand et al. 2026
Regulation and
License
Tight access rights can
Exclude rivals
market and
instrument dependent
Rule changes, new ones
Licenses
Weak for this case;
not an isolated legal case
checked
privileges according to
Acceptance not
available
None proven here
Moat
Porter as
Structural frame
network and
relationships
access to know-how,
execution partners,
Resources
All relationship changes, non-
Exclusivity
Tier 1: moderately indirect
for execution
Learning networks
accessible,
institutional deals
excluded
Relatively possible; absolutely
limited here
Anand et al. 2012;
Cohen/Levinthal
reputation and
Brand
Recruitment and

Raising capital
Indirectly all loss of reputation;
Effect depends on
Business model
Tier 3: weak for
Proprietary trading alpha
construction possible,
less immediate
Benefit
Sales month conceivable;
no direct signal moat
Helmer; Porter
Secrecy and
IP

Delays spread, protects
data and
Implementation details
All Independent Discovery,
functional replica,
Personnel changes
Tier 1: moderately indirect;
no patent effect
examined
Confidentiality
reachable
delay possible;
no guarantee against
Substitutes
McLean/Pontiff;
Barney

The taxonomy describes candidates, not ten proven moats. Expensive resources can fully show their added value as salaries,

Skim off license prices or financing costs. What matters is the company's remaining surplus. That's one economic derivation, the specific size of which remains open due to a lack of company information.

3 Question 1 and the horizon analysis

15 minutes

With short holding times, the cost burden can be large relative to the expected price movement. Microstructure effects also make this more difficult

Separation of tradable forecast and bid-ask bounce. Heston, Korajczyk and Sadka document half-hourly return patterns as well

short-term reversals; this is historical predictability, not a current 15-minute retail strategy. [Moderate for the transfers; Tier 1.] Heston et al. 2010.

As a separate conclusion, quote quality, executability, signal age and non-execution should be checked before model complexity. A

Models with a 15-minute forecast horizon can still have an edge that disappears seconds after signal generation.

Holding time, allowable execution latency and lifetime of a strategy are different time variables. For lack of direct

A fixed order "15 minutes always erodes faster than days" is not proven in a comparative study.

Several hours

Gao et al. provides historical evidence of intraday momentum: the first half hour forecast for the S&P 500 ETF examined including overnight component the last half hour; additional ETFs were included. This opens up hypotheses about the course of the day and

Information processing, but proves neither current net income nor exclusivity. [Moderate; individual empirical study with several tests.] Gao et al. 2018.

For this segment, the combination of few economically justified setups, selective participation and precisely measured appears to be the case

Execution plausible. This is a research prioritization, not a proven ranking. Institutional competition does not disappear because of this

a signal works outside of milliseconds.

One to several days

Lower necessary handling can reduce costs; However, holding time alone does not determine portfolio turnover. Novy Marx and

Velikov show the importance of cost reduction and different entry/halt thresholds for their anomalies. The examined monthly Turnover categories are not limits for multi-day strategies. [Strong cost findings in the original setting, moderate transfer.]

Novy-Marx and Velikov 2016.

On daily horizons, overnight jumps, financing, lending and, if necessary, futures rolls are added. Niche knowledge and more patient

Capital investments can be valuable; low trading frequency does not guarantee high signal content or a moat. As a work priority is

This segment is worth examining for limited resources if a concrete mechanism exists. The report does not establish a generally highest one

Retail success probability of this horizon.

4 Question 1 and Machine Learning

Why model performance and moat diverge

Gu, Kelly and Xiu show prediction gains of flexible ML methods in the empirical asset pricing task, including through nonlinear ones

Interactions. This is a positive finding about methods, not a test of the defensibility of a trading company. The work delivers in particular

no direct net live evidence for the short horizons asked here. [Moderate for ML utility; weak for retail moat.] Gu et al. 2020.

The fact that methods and code are generally available weakens the exclusivity of the architecture; it does not make all applications equal

good. The opposing position "Open source does not necessarily mean an advantage" would therefore also be unsubstantiated. Differentiation could be better

Task definition, time stamps, historical data states, rare domain knowledge and the interaction with execution arise.

[Own theoretical derivation from Barney and Cohen/Levinthal; direct evidence weak.]

Sculley et al. address data dependencies, technical debt, and problematic feedback loops in productive ML systems. Good

Model metrics alone are not enough for reliable operation. [Moderate technical evidence, not a trading moat study.] Sculley et al. 2015.

What could actually make your own data special

Proof of benefit module Additional moat test

Proprietary Labels Do they improve time-disaggregated forecasting or execution by cost? Are they difficult to reconstruct and not easily replaceable economically?

Own order history Does it improve estimates of fills, slippage and non-execution? Can a competitor collect the same information cheaply and quickly? Research memory Are repeated errors avoided and new hypotheses better prioritized? Does the advantage remain compared to a competent team with standard tools?

Adaptation infrastructure Does it reduce real errors, downtime or adaptation costs? Is this more than a generally available best practice?

Data Combination Does it provide incremental benefit over public baselines? Can alternative data produce the same economic benefit?

These tests are our own operationalization. A label that only calculates the next public return is through its internal storage not proprietary in the economic sense. A rare data field with no forecasting or execution utility is also not an asset.

Feedback is not an automatic self-reinforcer

Hagiu and Wright model data-based learning in competition. Whether this results in an advantage depends on the learning functions and

conditions of competition. Your customer data models do not support an automatic "Trade → Data → Alpha" spiral.

[Moderate theoretical support, weak trading transfer.] Hagiu and Wright 2023.

Own trades can generate execution knowledge, but they are selected: the model determines which situations are observed.

On the other hand, public subsequent returns from many non-traded situations can also be recorded without capital investment. Missing fills and not

chosen actions remain a counterfactual problem. Therefore, more orders do not produce proportionally more independent information. The

Improvement must be measured against a version without its own feedback data and over new time windows. [Own measurement logic; none

empirically determined retail learning curve.]

5 Question 2 and the analysis of H1

Theoretical foundation with a counterposition

Teece, Pisano and Shuen combine dynamic capabilities with coordination, specific resources and development paths. On trading

The idea fits when an organization truly renews its knowledge and inputs. Periodic retraining after consistent

Recipe is initially an operating routine. [Moderate, theoretical.] Teece et al. 1997.

Eisenhardt and Martin provide the central caveat: functionally similar capabilities may be achievable via different processes;

Advantages can lie in the resource configurations created as a result. It does not follow from H1 that regeneration would be useless,

but that their defensibility remains a separate proof. [Strong theoretical opposing position to equation, none empirical refutation of all trading moats.] Eisenhardt and Martin 2000.

The supplementary organizational literature specifies four mechanisms. Argote deals with the creation, storage and transmission of knowledge;

Cohen and Levinthal explain why prior knowledge influences the use of external knowledge. March shows the resource conflict between

Exploration and exploitation. Grant emphasizes the integration of specialized knowledge. [Moderate and indirect support for H1.] None of these works

provides a success rate for a retail edge factory. Argote, Cohen and Levinthal, March, Grant.

Alpha decay and contradictory findings

McLean and Pontiff examine 97 published return predictors. Portfolio returns are outside the original sample 26 percent and 58 percent below in-sample returns after publication. The authors interpret the difference in connection with publication-informed trading. These percentages are neither annual decay rates nor half-lives. [Strong for him examined average; Tier 1.] McLean and Pontiff 2016.

Jacobs and Müller examine 241 anomalies in 39 stock markets and find a reliable decline in publications across the country Main comparison for USA only. International barriers thus limit the generalizability of the US findings. [Moderate to strong than

geographical cross-check.] Jacobs and Müller 2020.

The replication issue isn't settled by "all data mining" either. Hou, Xue and Zhang find many insignificant anomalies

unified testing and a reduced role of microcaps. In contrast, Chen and Zimmermann demonstrate high reproducibility orientation towards the original definitions; Jensen, Kelly and Pedersen report additional international and Bayesian ones

Support for many factors. Different populations, weights, and tests explain some of the conflict; methodical

Reviews remain controversial. Reproduction does not mean today's profitability based on costs or exclusivity. [Strong for the existence of a methodological disagreement; moderate for individual generalizations.] Hou et al., Chen and Zimmermann,

Jensen

et al..

Performance persistence and the identification gap

Work Direct result What it doesn't show for H1

Carhart 1997 Factors, momentum and costs explain almost all of the fund persistence examined; particularly bad ones

Funds remain conspicuous

No proof that no one has skill; no observed discovery

Process

Kosowski et al.

2006

Bootstrap findings speak for skill and persistence of a minority of funds No separation of new edges from continued use of existing ones

Advantages

Fama and French

2010

Only a few funds cover their costs through benchmark-adjusted expected returns. Lack of investor net alpha is not the same as lack of it

Managerial ability

Work Direct result What it doesn't show for H1

Hounyo and Lin

2026

New wild bootstrap method finds outperforming funds; historical outperformance focuses loudly

Publisher's presentation before 2003

Methodological reassessment is not retail ML proof of regeneration

Anand et al. 2012 Institutional Trading Desks Show Relative Persistence in Execution Performance Execution competence is not the repeated discovery of new ones

Prognostic signals

Evidence rating: moderate and heterogeneous for skill, weak for explicitly identified alpha regeneration. Sources: Carhart, Kosowski

et al., Fama and French, Hounyo and Lin, Anand et al..

Berk and Green show theoretically how capital inflows and diminishing returns to scale can lead to persistence of returns despite differences

Managerial ability can be eliminated. Pástor, Stambaugh and Taylor provide evidence of scale erosion at the industry level: improved skill works

does not automatically correlate with improved fund performance. Their bias-adjusted individual fund economies of scale are not

significant. [Moderate: model plus observational empirical evidence; no causal retail evidence.] Berk and Green, Pástor et al..

Four partial hypotheses and separate verdicts

Hypothesis Verdict Confidence and Justification

H1a Real measurable ability Partially supported Moderate for organizational learning differences; low for the specific alpha regeneration measurement. Prospective

Cohorts can make the construct testable.

H1b Repeats over-random

Regeneration

Not sufficient in this precise form supported

Moderate confidence in the identified evidence gap, no statement about impossibility. Skill/persistence studies predominantly observe results rather than new edge generations.

H1c Can be built up through retail Partially supported as technical and organizational possibility

Low to moderate confidence. A tight process can be realistically designed; its profitable production rate is not documented.

H1d Against competent

Copycats defensible

Not supported for the standard case; conditional plausible

Moderate confidence in the lack of a general guarantee. Non-substitutable inputs or cumulative, economic relevant knowledge would have to be added.

The confidence statements are reasoned qualitative judgments, not calculated posterior probabilities. H1 as a conjunction of all four

Partial hypotheses have not been proven for the actor described. The overall label "partially supported" refers to the conditional

Mechanism, not on the claim of an already existing company with proven moat.

Necessary and sufficient

The production chain to be tested is: data, knowledge, compute, capital and live feedback → hypotheses and tests → selected strategies → real execution → additional net returns and new knowledge. There is still an output "validated backtest".

no economic output. The process chain is its own analytical representation, not an empirically estimated model in this form.

With a portfolio of ephemeral edges, replacement, improvement or another source of income must emerge in a timely manner if the

Trading activity should continue. It does not follow that completely new signals are universally necessary. Unchanged structural

Rewards, recurring opportunities, and adjustments to existing strategies may suffice; an orderly exit is also one rational alternative. An information advantage can end even though the research process continues to work well.

For a moat, even repeated success alone is not enough: equally good competitors could have the same success with different code and

achieve comparable costs. H1 therefore additionally needs an economic barrier against imitation or substitution. A cumulative feedback loop is a possible, but not necessarily the only barrier; exclusive rights or scarce something difficult to replace

Knowledge could carry alternatively. [Own synthesis; moderate theoretical foundation.]

6 conditions for a real regeneration moat

When H1 could become viable

1. Economic benefits: New generations of strategies generate incremental risk-adjusted net returns; additional research and Operating costs are covered.

2. Repeatability: Several predefined, time-separated research cohorts work; Failed attempts remain in the denominator.

3. Timeliness: The remaining economic useful life after development and validation is sufficient for a positive outcome expected life cycle value.

4. Barrier: Relevant inputs, knowledge or processes are costly, slow or functionally difficult for competent competitors replace.

5. Appropriation: The added value remains with the company and is not completely passed on to data providers, employees or

Financing partner provided.

6. Capacity and survival: Actual allocable capital as well as common liquidity and drawdown risks are taken into account.

7. Learning transfer: Findings demonstrably improve new tasks; they not only reflect more favorable market phases.

When the claim fails

1. Commodity search: Equivalent public data and standard procedures produce the same output similarly quickly and cheaply.

2. Erosion overtakes production: Discovery, validation and deployment take longer than the exploitable opportunity, or new ones

Net income does not replace expiring income.

3. No cumulative advantage: More history does not improve forecast or execution; without any other barrier, the only thing left is running

Effort.

4. No imitation barrier: Secrecy protects specific files, but not the economic function.

5. Selection bias: Only winners, surviving companies or successful variants are recorded.

6. Hidden risks and costs: The supposed alpha is factor, liquidity or tail risk or disappears after Execution and fixed costs.

These are test conditions derived from the literature, not empirically calibrated success thresholds. Its theoretical basis lies especially in Barney, Teece et al., Eisenhardt/Martin, Berk/Green and the literature on multiple testing.

7 Retail reality check and the small Edge Factory

Attainable and unattainable benefits

According to the order, co-location advantages, privileged order flow, exclusive feeds and institutional prime broker conditions are no longer applicable.

Achievable are narrow domain knowledge, error prevention, clean historical data, custom execution protocols and the freedom to

to avoid unsuitable trades. First of all, these are skills and scope for creativity. Their mere accessibility does not prove positive

Expected value.

Barber et al. examine day traders in Taiwan from 1992 to 2006. Less than one percent of the population earns using their methodology

predictable and reliable positive net abnormal returns. This disproves "retail can never have skill," but it doesn't justify it

Probability of success for today's ML traders or the conclusion that the winners have constantly invented new signals.[Strong within the

historical study, moderate to weak for today's transfer.] Barber et al. 2014.

Anand et al. examine retail limit orders based on over 27 million orders with 19 US retail brokers in May 2020. Patient limit orders

have lower costs; the measurement also takes into account the opportunity costs of unexecuted quantities. That is relevant positive

Evidence for retail execution. The short, special examination period and the selection of differently motivated orders limit a causal generalization. A time-critical ML signal can lose its usefulness if it waits for a long time. [Moderate for this one considered setting.] Anand et al. 2026.

The advantage of small size does not automatically protect a niche

Small orders may have lower impact of their own, and limited absolute profit may be economically relevant to an individual be. However, institutions can take small positions or aggregate many small opportunities. Small size is also common imitable. Only when the niche remains attractive after all costs and competition is permanently limited for economic reasons does it arise

more than a scaling advantage. [Own derivation; Capacity mechanism moderate, moat barrier weakly documented.]

Institutional cost evidence should not be used across the board against retail: Frazzini, Israel and Moskowitz find real ones institutional trades significantly lower costs than some previous estimates. But your data comes from an individual institutional managers and do not provide the same retail conditions. [Tier 3, moderate methodological substance and limited external validity.] Frazzini et al. 2018.

How small the factory can be

A narrow pipeline for a market mechanism is organizationally plausible. Their minimal functionality includes data origin, Hypothesis register, reproducible test, cost model, small prospective validation and operational monitoring. An individual actor can

Completing these tasks sequentially carries concentration, time and key person risks. An internally referred to as an "organization".

Collection of scripts does not address these risks. [Own organizational recommendation; no documented optimal team size.]

The honest answer is: A relative quality advantage over undisciplined retail is plausible; a robust one

Moat compared to competent retail teams and institutions is usually difficult and cannot be proven here. A not

Protected edge can still be profitable. Therefore, neither the label "moat" nor the lack of it is sufficient for a business thesis;

The decisive factors are life cycle surplus and the consciously accepted risk of erosion.

8 negative list

More parameters or deep learning: Model complexity alone does not create a resource that is difficult to replace.

Lots of backtests: More search attempts increase the number of choices and thus the selection problem; they are not

Proof of productivity.

High in-sample sharpness: Insufficient without fully recorded trials, costs and independent future verification.

Profit without risk decomposition: market, factor, liquidity or catastrophe exposure can appear as a skill.

Illiquid stocks: Larger price deviations can be a reflection of high trading costs and risks.

Private repositories: Hide code but do not prevent independent discovery of the same economic idea.

Automatic retraining: Operational automation demonstrates neither adaptability nor profitable learning transfer.

Many own trades: Activity is not a measurement of independent new information.

Known successful quant companies: Without an accessible complete strategy and failure history, they are not proof of the concrete retail hypothesis.

"We are always finding new edges": Without prospective cohorts, life cycle calculations and benchmarks, this is a hope not a verified company resource.

Evidence: The selection criticism is strongly theoretically and methodologically based; the concrete list is our own synthesis.

Bailey et al. develop

PBO/CSCV to study backtest selection. Harvey, Liu, and Zhu demonstrate the consequences of multiple factor testing. Not a single one

Diagnostic procedures guarantee future profitability. Bailey et al., Harvey et al.

9 Prioritized Implications for Retail Quant Lab

Order of construction

Priority Concrete work Decision criterion

1 Economics

define

Establish a universe, instruments, eligible orders, risk and cost benchmark;

Evaluate time expenditure

Is there a plausible profit mechanism, the magnitude of which costs and

Insecurity justified?

2 Data and validation Availability at the time of decision, corporate actions, delistings, quotes and all

Versioning research experiments

The result remains the same if the test is carried out in a timely, cost-realistic manner

3 Measure execution decisions, orders, changes, fills, aborts and unexecuted quantities

capture

Prospective cost assumptions are consistent with observed results

4 Deepen narrow knowledge Test a small number of economically related hypotheses; Simple baselines retain additional benefits beyond the known exposure and baseline

5 Test regeneration Freeze old production version, define new cohorts in advance and compare fairly New generation improves the overall net return with the same risk budget

6 Test defensibility Define competent replication with public inputs and the same budget as a comparison Advantage remains economically significant and a plausible barrier explains it

This is our own evidence-informed recommendation, not a study about an optimal roadmap. Proprietary data collection should first

be expanded when ablation shows its additional benefit. Additional compute spending follows a proven bottleneck, not a blanket assumption that larger models are more profitable.

Measuring system for H1a to H1d

Metric definition and protection against self-deception

Research cohort All hypotheses registered before the start of a time window including rejected approaches; Mark variants of the same idea as a family

Qualified discovery rate Number of commercially confirmed new strategy families per research time or budget; Show uncertainty and immature cohorts separately

Incremental live contribution New overall process minus frozen comparison version, with the same capital/risk budget and after execution and additional operating costs

Metric definition and protection against self-deception

Time to usability hypothesis to deployment separated into data work, testing and operation; don't just measure computing time

Life cycle value Realized net income plus conservatively estimated residual value minus development/operation; Forecast and realization separately

Erosion and replacement performance Expiring expected net income and additional contribution from new cohorts on the same time/capital basis; Don't compare mere strategy numbers

Learning transfer Same task with and without your own history/labels; comparable data budgets and time-based tests

Disadvantages of imitation time, costs and performance of a competent functional replica; identical code is not required

Dependency Proportion of the result from a few market phases, factors, strategy clusters and a key person

A sensible internal target is additional economic value per research cohort, not "more successful backtests". At

of an allocation change, joint trades and net orders must be treated consistently, otherwise diversification or

Cost netting counted twice. A moat comparison also needs competitors: an improvement over your own old version measures learning ability, but not yet relative defensibility.

Prospective test design

Before each cohort, the mechanism, data access, model choice space, target size, cost assumptions and termination rules are recorded.

Development and later evaluation are separated in time; In the case of overlapping labels, information overlaps must be removed and

appropriate time intervals must be taken into account. Repeated readjustment after viewing the test window turns this window into a

Development sample. A simple mechanistic approach, the frozen production version and a resource-equal public version serve as a comparison

pipeline. Paper trading checks processes, but only incompletely measures real fills and impact. A limited live test serves the purpose

execution validation; the size must follow from the previously accepted risk. Statistics are based on appropriate time blocks

and shared

Market exposure coordinated. Thousands of correlated trades do not replace independent market phases.

Multiple testing corrections and PBO are complementary diagnoses. Neither a blanket t-level nor a fixed number of months is considered

universal release limit used. The duration required depends on effect size, dependence, cost uncertainty and regime coverage.

[Own experimental setup, methodically supported by Bailey et al. and Harvey et al.]

Falsification rule: Does the additional economic contribution via sufficiently informative prospective cohorts remain insufficient or is it

reproducible through an equivalent public pipeline, H1 should be downgraded for this company. The result can be “more useful

“Operating process without moat”. Failure does not automatically justify more searching; Market segment, database or continuation of the

Business models must then be re-evaluated.

10 research gaps

The biggest gap is a prospective panel of research organizations with complete hypothesis, failure, deployment and

Cost histories. Fund returns alone cannot identify the regeneration of new signals. Also missing for the specific setting

Reliable learning curves of your own retail order data, a separation of organizational performance and personal talent as well as controlled

Comparisons of Competent Public and Proprietary Pipelines.

Other open questions concern the current capacity of small niches, execution quality for short-lived ML signals, transmission to

Futures/FX/Crypto and the change through broad tool access. How quickly a proven knowledge advantage can be functionally substituted

is also open. The search did not identify any sufficiently direct studies on this; this is not a claim that outside of the checked source space none exist.

For own empirical work have full cohort registries, coupled live/baseline protocols and ablations of data feedback

Priority. Success would not only be a significant difference in returns, but also a positive economic impact across several tasks

understandable barrier. Without a competitor comparison, even a good learning experiment remains a test of ability and not moat.

What practice believes and science has not yet proven

A r/quant thread from May 2026 discusses whether small capacity-constrained strategies keep institutions out or through automation

but become attractive. Both positions appear there. Unverified practitioner anecdote (Reddit, Tier 4); not representative

Consensus measurement. The academic capacity and cost literature supports possible differences in scale, not an exclusive retail entitlement

on niches. Discussion on low-capacity markets.

A r/algotrading thread questions the usefulness of public data and simple scripts and discusses the niche argument.

Unverified practitioner anecdote (Reddit, Tier 4). The reviewed literature justifies skepticism over unvalidated ones

Promise of success, but it also does not prove that “every public strategy is worthless”. There is no success rate from such threads

win another industry verdict. Discussion about retail alpha.

11 Complete bibliography by level of evidence

All following sources were researched on September 8, 2026. “Full text” means that the original text or a

Author's version was accessible and relevant sections were checked; it does not mean independent replication of all calculations.

Journal year and online/working paper year are separated if necessary. Short references mentioned in the text refer to these entries.

Tier 1 peer review and academic monographs

Anand, Amber; Irvine, Paul; Puckett, Andy; Venkataraman, Kumar (2012). Performance of Institutional Trading Desks: An Analysis of Persistence in Trading Costs. *The Review of Financial Studies* 25(2), 557–598. Empirical observational study of institutional

transactions; Abstract and bibliographic verification, author PDF found. DOI 10.1093/rfs/hhr110.

Anand, Amber; Samadi, Mehrdad; Sokobin, Jonathan; Venkataraman, Kumar (2026; online 2025). Retail limit orders.

Review of Finance 30(2), 459–488. Empirical retail order study; Open access full text, sample and cost measurement checked. DOI

10.1093/rof/rfaf049.

Argote, Linda (2013). *Organizational Learning: Creating, Retaining and Transferring Knowledge*. 2nd edition, Springer.

Academic

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Barber, Brad M.; Lee, Yi Tsung; Liu, Yu Jane; Odean, Terrance (2014). The cross-section of speculator skill: Evidence from day trading. *Journal of Financial Markets* 18, 1–24. Empirical study, Taiwan 1992–2006; Publisher's abstract and detailed introduction checked. DOI 10.1016/j.finmar.2013.05.006.

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Eisenhardt, Kathleen M.; Martin, Jeffrey A. (2000). Dynamic Capabilities: What Are They? *Strategic Management Journal* 21(10-11), 1105-1121. theory/literature synthesis; public original PDF checked, especially pages 1106 and 1117. DOI 10.1002/1097-0266(200010/11)21:10/11<1105::AID-SMJ133>3.0.CO;2-E. Original text.

Fama, Eugene F.; French, Kenneth R. (2010). Luck versus Skill in the Cross-Section of Mutual Fund Returns. *The Journal of Finance* 65(5), 1915-1947. Empirical bootstrap study; Publisher's abstract and bibliographic information checked. DOI 10.1111/j.1540-6261.2010.01598.x.

Gao, Lei; Han, Yufeng; Li, Sophia Zhengzi; Zhou, Guofu (2018). Market intraday momentum. *Journal of Financial Economics* 129(2), 394-414. Empirical intraday study; University publication page with abstract checked. DOI 10.1016/j.jfineco.2018.05.009.

Grant, Robert M. (1996). Toward a Knowledge-Based Theory of the Firm. *Strategic Management Journal* 17(S2), 109–122. theory articles; Publisher source checked. DOI 10.1002/smj.4250171110.

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Hagiu, Andrei; Wright, Julian (2023). Data-enabled learning, network effects, and competitive advantage. *The RAND Journal of Economics* 54(4), 638–667. Theoretical competition model; Publisher's abstract and author's version checked in December 2021. DOI 10.1111/1756-2171.12453.

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Heston, Steven L.; Korajczyk, Robert A.; Sadka, Ronnie (2010). Intraday Patterns in the Cross-Section of Stock Returns. *The Journal of Finance* 65(4), 1369-1407. Empirical microstructure study; Publisher's abstract and author's preprint checked. DOI 10.1111/j.1540-6261.2010.01573.x, arXiv 1005.3535.

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Finance 78(5), 2465–2518. Bayesian replication analysis and international data expansion; Publisher/institute abstract and Results sections checked. DOI 10.1111/jofi.13249.

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Porter, Michael E. (1980). Competitive Strategy: Techniques for Analyzing Industries and Competitors. Free Press. Research based

Strategy monograph, not an empirical retail test; Checked bibliographically and via explanation from the author institute, not a complete one

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Tier 2 working papers and institutional versions

The following accessible versions were used to examine journal papers already listed above. They do not count as additional independent evidence: Chen/Zimmermann (2021), Open Source Cross-Sectional Asset Pricing, FEDS 2021-037, DOI 10.17016/FEDS.2021.037; Gu/Kelly/Xiu (2018, revised 2019), NBER WP 25398, DOI 10.3386/w25398; Harvey/Liu/Zhu (2014), NBER WP

20592, DOI 10.3386/w20592; Pástor/Stambaugh/Taylor (2014), NBER WP 19891, DOI 10.3386/w19891. Unpublished newer hits without

Necessary additional benefits were not included in the verdict.

Tier 3 Practitioner Research and Monographs

Frazzini, Andrea; Israel, Ronen; Moskowitz, Tobias J. (2018). trading costs. AQR Working Paper, August 23rd. Empirical Practitioner research based on live institutional execution; Research page/abstract checked. Despite the working paper title here due to Practitioner data origin and usage conservative Tier 3. Original source.

Helmer, Hamilton (2016). 7 Powers: The Foundations of Business Strategy. Deep Strategy. ISBN 9780998116303.

Practitioner

monograph; Author website and original definitions checked, not full book access. Author website, definitions at Strategy Capital.

Tier 4 Hypothesis Generating Forum Discussions

Anonymous Reddit Posts (2026). Is it a myth that retail has an edge over institutions in significantly less liquid markets/lower capacity

strategies? r/quant, from May 4th. Unverified practitioner anecdotes; Checked thread excerpts. thread.

Anonymous Reddit Posts (2026). Is alpha even real for retail at this point or are we all just deluding ourselves. r/algorithmic. Unverified practitioner anecdotes; Indexed thread excerpt checked, current date not verified separately. thread.

Uncertainties and limitations of this research

The attachment to the order does not contain any real company, capital, data or broker information. All company-specific statements are

therefore conditional derivations under the stated standard assumptions. The source material is heavy on historical stock and Fund samples concentrated. Monthly studies cannot be treated as direct tests of the effectiveness of 15-minute strategies. Relevant original texts and publicly available abstracts/previews were reviewed; not all books and journal full texts were accessible. Individual publisher requests failed, which is why author or institutional versions were used. The numbers in this Reports are limited to visible evidence; no separate replication of the data analyzes was carried out. Source prestige and peer

Reviews do not eliminate model dependency or publication and selection problems.

The search was not fully reproducible across licensed scientific databases and provides no guarantee of completeness on the research deadline. It was ended after all the required topic areas, central opposing positions and the remaining ones Identification gaps were recorded. Further general literature searches do not replace the missing prospective organizational data.

The reliable result is therefore conditional: regeneration can enable or support a moat if its economic effect has been repeatedly proven and its underlying advantage is difficult to replace. For Retail Quant Lab, both are initially to demonstrate.

Source-based synthesis · Standard assumptions and evidence limits identified in the report